EXPERIMENT 4

DATA MODELLING AND FORM DESIGN FOR A BUSINESS APPLICATION USING ZOHO CREATOR

1. Aim

To study and document data modelling and form design in Zoho Creator using the existing Parking Slot Booking System.

2. Software Requirements

- Zoho Creator
- Google Chrome web browser
- Microsoft Word

3. Theory

Data modelling is the process of identifying, organising and structuring the data required by an application. A well-designed data model helps maintain consistency, minimise duplicate data and support efficient data retrieval.

Zoho Creator is a low-code application development platform that enables users to create forms, reports, workflows and dashboards through a visual development environment.

Forms are used to collect and store information, reports display and organise stored records, and dashboards provide a central view of application components and information.

The Parking Slot Booking System uses these concepts to manage vehicle owner details, parking slot information and booking-related activities.

4. Application Description

The Parking Slot Booking System is designed to organise vehicle owner information, parking slot details and parking bookings. The existing application contains the following components:

1. Vehicle Owner
2. Vehicle Owner Report
3. Parking Slot
4. Parking Slot Report
5. Booking

6. Dashboard

5. Procedure

Step 1: Access the application

Opened the existing Parking Slot Booking System in Zoho Creator and accessed its application design interface.

Step 2: Examine the Vehicle Owner form

Inspected the Vehicle Owner form, which contains the following fields:

- Owner ID
- Full Name
- Phone Number
- Email
- Vehicle Number
- Vehicle Type
- Department

These fields are used to organise information about vehicle owners and their vehicles.

Step 3: Examine the Parking Slot form

Inspected the Parking Slot form, which contains the following fields:

- Slot Number
- Floor
- Slot Type
- Status

These fields represent the identification, location, type and status of parking slots.

Step 4: Identify reports

Identified the Vehicle Owner Report and Parking Slot Report in the application navigation. These reports provide access to the corresponding records.

Step 5: Examine the Booking module

Identified the Booking module, which is intended to manage parking booking information.

Step 6: Examine the Dashboard

Identified the Dashboard component, which provides a central navigation point for the application.

Step 7: Review the application structure

Reviewed the available forms, reports and modules to understand how the application organises its information and supports parking management.

6. Data Model

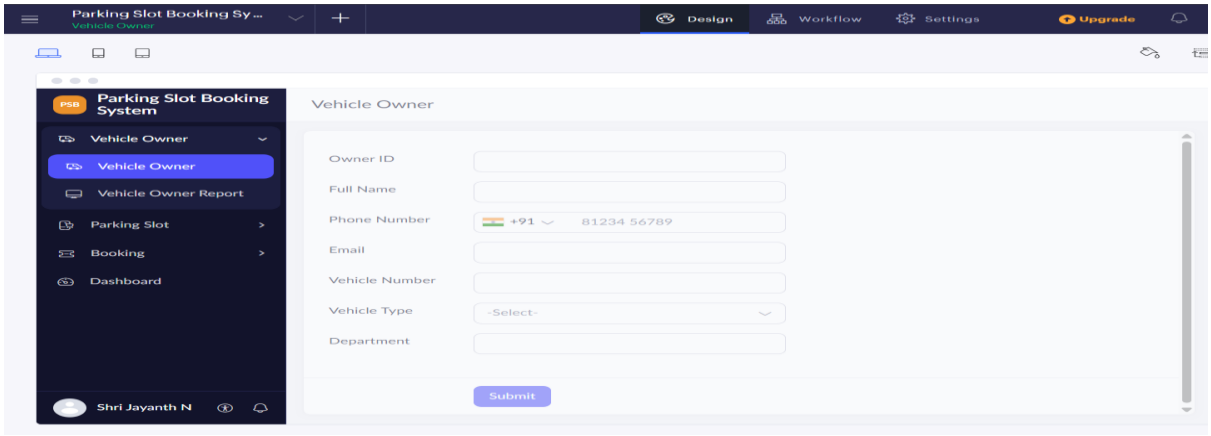
The application can be represented using the following conceptual data model:

- **Vehicle Owner:** Stores owner identification and contact details.
- **Parking Slot:** Stores slot identification, floor, slot type and status.
- **Booking:** Represents a parking reservation associated with a vehicle owner and a parking slot.

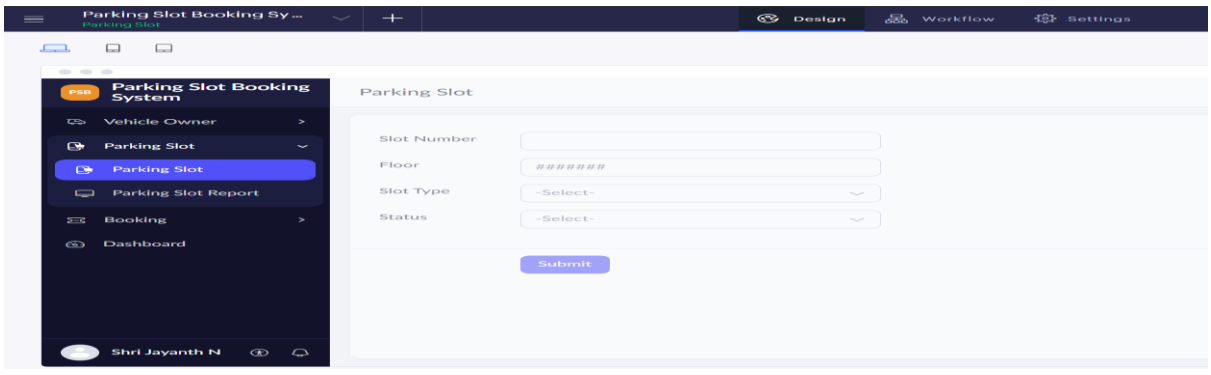
Conceptually, one vehicle owner may have multiple bookings, and one parking slot may be associated with different bookings over time. The actual relationships depend on the fields and lookup configurations implemented in the application.

7. Screenshots

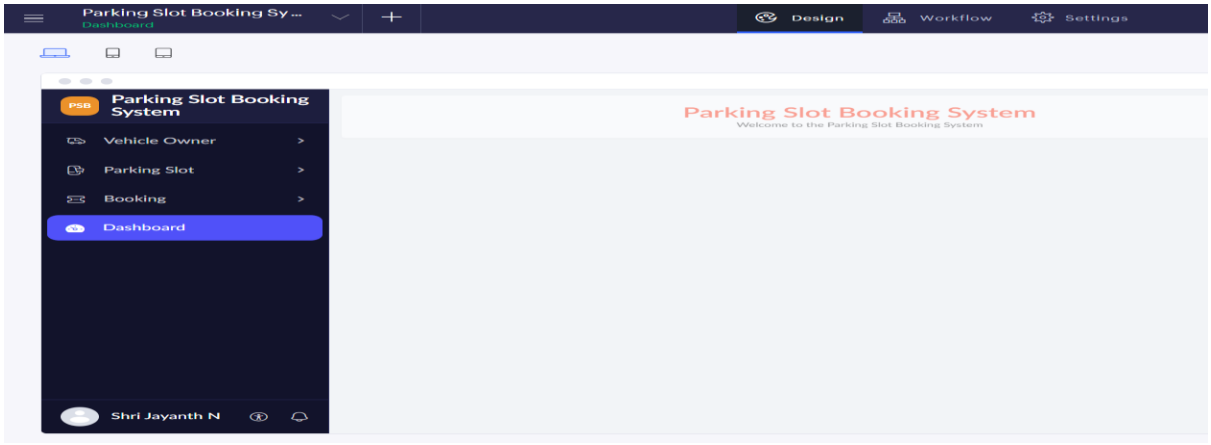
Screenshot 1: Vehicle Owner Form



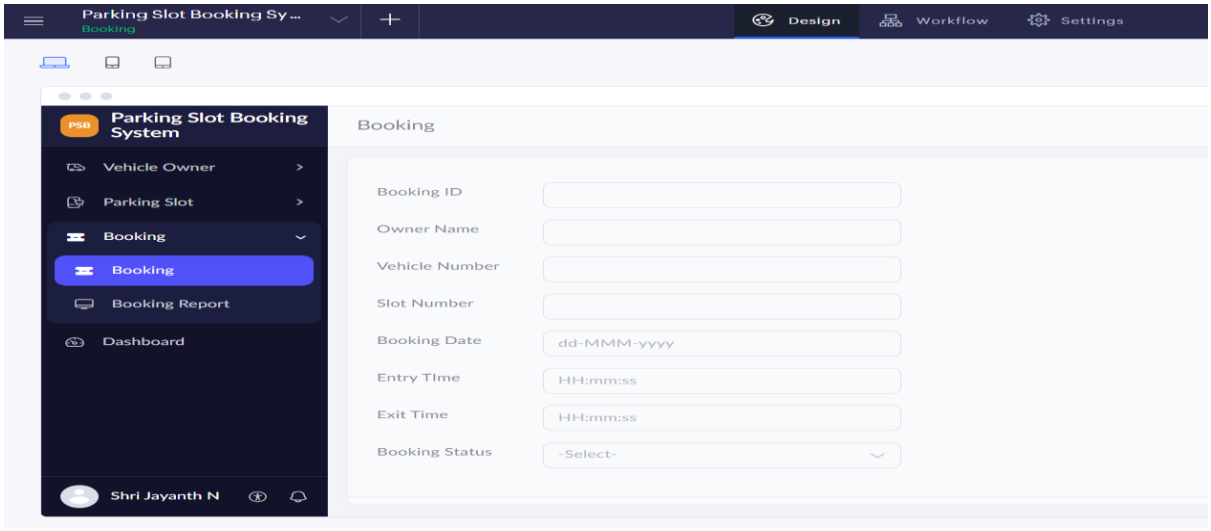
Screenshot 2: Parking Slot Form



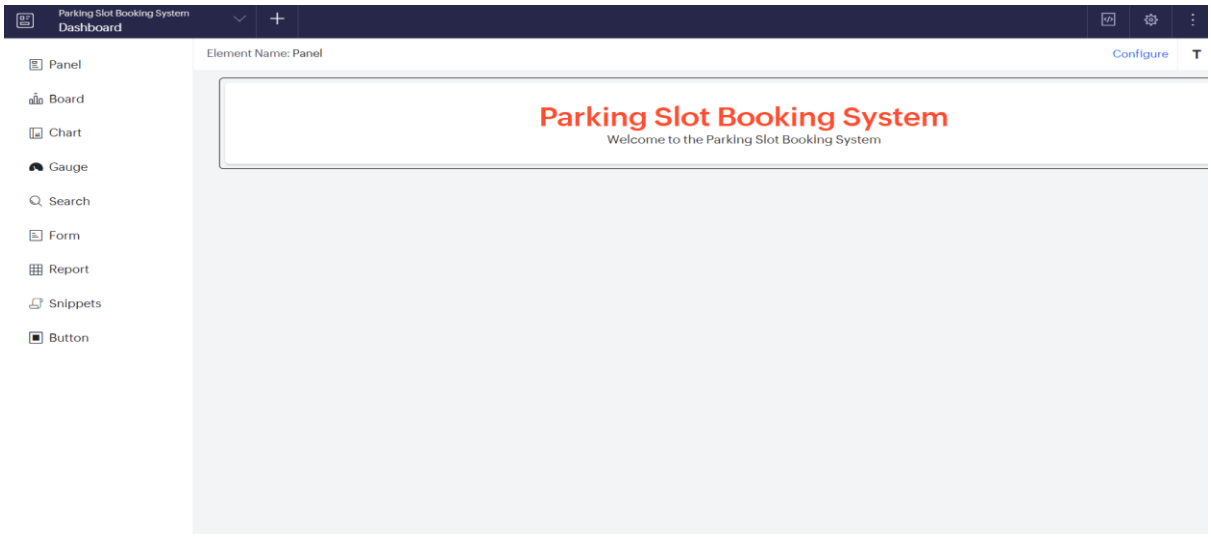
Screenshot 3: Application Navigation



Screenshot 4: Booking Form



Screenshot 5: Dashboard



8. Result

The existing Parking Slot Booking System was examined in Zoho Creator. Its Vehicle Owner and Parking Slot forms, associated reports, Booking module and Dashboard were identified and documented to demonstrate the concepts of data modelling and form design in a low-code application.

9. Conclusion

The experiment provided an understanding of how data is organised using forms and conceptual entities in Zoho Creator. The existing Parking Slot Booking System demonstrates the use of a low-code platform to structure information and provide a foundation for parking management.